FEDERAL BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, ISLAMABAD

INTERMEDIATE (11TH CLASS) EXAMINATION, 2025

PHYSICS

Past Paper 2025 • Paper Code: 1HA-I 25004 — 30041 (Version B) / 32041 (Version D)

Retyped faithfully from the original board paper — same questions, no changes.

VERSION B

PHYSICS HSSC-I — SECTION – A (Objective Type)

Time Allowed: 25 Minutes | **Maximum Marks:** 17 | **Paper Code:** 30041 (Version B)

Note: Section – A is compulsory. All parts of this section are to be answered on this page and handed over to the Centre Superintendent. Deleting/overwriting is not allowed. Do not use lead pencil.

حصہ اول لازمی ہے۔ اس کے جوابات اسی صفحہ پر دے کر ناظم مرکز کے حوالے کریں۔ کاٹ کر دوبارہ لکھنے کی اجازت نہیں ہے۔ لیڈ پنسل کا استعمال ممنوع ہے۔

Note: Fill the relevant bubble against each question according to curriculum:

ہر سوال کے سامنے دیے گئے کریکولم کے مطابق درست دائرہ کو پر کریں۔

Q1. The dimensions $[ML^2T^{-2}]$ show:

(A) Power
(B) Work
(C) Velocity
(D) Acceleration

Q2. If r = 2.25±0.01cm then percentage uncertainty in "r" will be:

(A) 0.2%
(B) 0.3%
(C) 0.4%
(D) 0.5%

Q3. $\hat{j}.(\hat{i}\times\hat{k})$ =

(A) 0
(B) −1
(C) 1
(D) 2

Q4. Magnitude of centripetal force Fc =

(A) $mr\omega^2$ (B) $mr\omega$

(C) $m\omega^2$ (D) mv^2

Q5. If the length of blade of a fan is increased, then its rotational speed (at same voltage and current):

(A) Will increase (B) Will decrease

(C) Will become infinite (D) Remains constant

Q6. Two bodies A and B of same masses have velocities in the ratio 2:1, their kinetic energies are in the ratio:

(A) 2:1 (B) 1:2

(C) 4:1 (D) 1:4

Q7. Maximum drag force on a 1kg falling sphere is:

(A) 9.8 N (B) 1 N

(C) 98 N (D) 4.9 N

Q8. Young's Modulus, Y=

(A) $(\Delta L/L)/(F/A)$ (B) $(FA)/(\Delta L.L)$

(C) $(\Delta L.L)/(FA)$ (D) $(F/A)/(\Delta L/L)$

Q9. In an adiabatic process which quantity remains constant?

(A) Internal energy (B) Volume

(C) Temperature (D) Mass

Q10. Location of a submarine in deep sea can be detected by:

(A) Doppler effect (B) Compton's effect

(C) Faraday's Law (D) Bernoulli principle

Q11. An electron is placed in a uniform electric field directed from south to north. In which direction will electron move after it is released?

(A) East (B) South

(C) North (D) West

Q12. In order to achieve high accuracy, the slide wire of a potentiometer should be:

(A) As long as possible (B) As short as possible

(C) Very thick (D) Neither too small, nor too large

Q13. Two closely placed parallel wires are carrying current in opposite direction, the wires will:

(A) Repel each other (B) Attract each other

(C) Not affect each other (D) Rotate each other

Q14. One weber is equal to:

(A) NA^{-1} (B) NmA^{-1}

(C) $Nm^{-1}A$ (D) $Nm^{-1}A^{-1}$

Q15. When an object moves with a very high speed its length _____ in the direction of its motion.

(A) Contracts (B) Expands

(C) Remains same (D) Becomes zero

Q16. Relativistic mass is always:

(A) Less than rest mass (B) Greater than rest mass

(C) Equal to rest mass (D) Zero

Q17. A hadron may either be a Baryon or a:

(A) Lepton (B) Gluon

(C) Photon (D) Meson

> Note: Important Formulae: $\bar{A}.\bar{B} = ABCos\theta$ • $L = I\omega$ • $K.E = \frac{1}{2}mv^2$ • $W = mg$ • $g = 9.8ms^{-2}$ • $v = r\omega$ • $\bar{A}\times\bar{B} = ABSin\theta$ • $\varphi_B = \bar{B}.\bar{A}$ • uncertainty = Absolute uncertainty/No.of vibrations

SECTIONS – B & C (Subjective Type)

Time Allowed: 2:35 Hours | **Maximum Marks:** 68 | **Paper Code:** 1HA-I 25004 (B)

Q2. (Marks (14 x 3 = 42)) Q.2 Answers the following parts briefly.

(i) [1x3] Briefly explain 'precision' and 'accuracy' with a diagram. — OR (03) — How can ordinary light be polarized using polarizers?

(ii) [03] The velocity of a missile traveling at 1500 ms^{-1} increases to 5000 ms^{-1} after moving through a distance of 70 km. Calculate the time for the missile to reach this velocity. — OR (03) — A body at 200°C undergoes a reversible isothermal process. The heat energy removed in this process is 7300 J. Calculate the entropy change and discuss entropy sign.

(iii) [03] What is meant by 'conservative' and Non-conservative' forces? Give one example each. — OR (03) — What is meant by Magnus effect? Explain with the help of an example.

(iv) [03] Differentiate between crystalline solids and amorphous solids. (Give at least three differences.) — OR (1+2) — Earth completes its one rotation about its axis in 24 hours. Calculate angular velocity of earth about its axis in radian per second.

(v) [03] The Doppler's effect is used in radar system. Briefly explain the working of radar system. — OR (03) — Verify that: $(\bar{A}.\bar{B})^2 + (\bar{A}\times\bar{B})^2 = (AB)^2$

(vi) [03] Calculate KE in electron volt and joules gained by an electron moving through a potential difference of 2 volts. — OR (03) — What happens to the density of an object as it moves in a direction with very high speed relative to speed of light? Justify.

(vii) [03] Why does the resistance of LDR decrease with increase in intensity of light on it? — OR (2+1) — State Faraday's law of electromagnetic induction. Give its mathematical form.

(viii) [1.5+1.5] Does the induced emf in a circuit depend on the resistance of the circuit? Does the induced current depend on the resistance of the circuit? Justify. — OR (03) — At what speed would the mass of proton be doubled? The rest mass of proton is 1.673×10^{-27} kg.

(ix) [03] The Kinetic energy of a charged particle moving in a uniform magnetic field does not change. Why? Explain briefly. — OR (03) — If $^{233}_{92}U$ decays twice by alpha particle emission, what is the resulting isotope? Show by equation.

(x) [03] What is Faraday's cage? Briefly explain its working. — OR (03) — Compare gluons and photons.

(xi) [03] What is meant by annihilation? Give one annihilation reaction. — OR (03) — To increase the intensity of wave by a factor of 10, calculate the required increase in amplitude of wave.

(xii) [03] How will the radius of a flexible ring change if it is given negative charge? — OR (03) — What is the difference between inertial and non-inertial frame of reference?

(xiii) [03] Use dimensional analysis to prove the equation $P = \rho gh$ is dimensionally homogeneous. — OR (03) — Write three statements of second law of thermodynamic. (one in terms of entropy)

(xiv) [1x3] Derive the relation between torque, moment of inertia and angular acceleration. — OR (1+2) — What is meant by interference? Give conditions for interference.

> Note: Attempt the following questions. (Use of graph paper is not necessary. Candidates can make their own grid on answer book if required)

Q3. (Marks 3+4) Q.3 State and explain vector product with an example. Also show that $\bar{A}\times\bar{B} \neq \bar{B}\times\bar{A}$ — OR (1+2+4) — What is meant by projectile? Derive the mathematical expressions of following for projectile motion: i) Time of flight ii) Range and maximum range of projectile

Q4. (Marks 4+2) Q.4 Explain the working of refrigerator. Also explain the coefficient of performance of refrigerator. — OR (06) — State work-energy theorem. Explain this theorem for resistive medium.

Q5. (Marks 1+5+1) Q.5 State the equation of continuity and derive its mathematical form. On what basis is the equation of continuity derived? — OR (1+6) — What is meant by elastic potential energy? Find a relation for the elastic potential energy of a material.

Q6. (Marks 1+1+4) Q.6 Why does an astronaut feel weightless in space? How can artificial gravity be produced in a satellite? For a satellite derive expression for

angular velocity, time period and frequency of satellite. — OR (06) — Find the current flowing through each resistor of the circuit.

> Note: [A circuit diagram (10V battery, 2Ω resistor, 6V battery, 1Ω resistor) is printed next to Q.6 OR in the source scan]

> Note: Important Formulae: $\text{Sin}^2\theta+\text{Cos}^2\theta=1$ • $\omega=\theta/t$ • $S=\Delta Q/T$ • $1eV=1.6\times10^{-19}J$ • $\bar{A}.\bar{B}=ABCos\theta$ • $\sum\varepsilon=0$ • $\omega=2\pi/T$ • $\bar{A}\times\bar{B}=ABSin\theta$ • $E=qv$ • $I=\varepsilon/R$ • $vf=vi+at$ • $vf^2-vi^2=2aS$ • $Fc=mv^2/r$ • $\varepsilon=\Delta6/\Delta t$ • $I\propto A^2$ • $\sum I=0$ • $m=m_0/\sqrt{(1-v^2/c^2)}$

VERSION D

PHYSICS HSSC-I — SECTION – A (Objective Type)

Time Allowed: 25 Minutes	**Maximum Marks:** 17	**Paper Code:** 32041 (Version D)

> Note: Section – A is compulsory. All parts of this section are to be answered on this page and handed over to the Centre Superintendent. Deleting/overwriting is not allowed. Do not use lead pencil.

> حصہ اول لازمی ہے۔ اس کے جوابات اسی صفحہ پر دے کر ناظم مرکز کے حوالے کریں۔ کاٹ کر دوبارہ لکھنے کی اجازت نہیں ہے۔ لیڈ پنسل کا استعمال ممنوع ہے۔

> Note: Fill the relevant bubble against each question according to curriculum:

> ہر سوال کے سامنے دیے گئے کریکولم کے مطابق درست دائرہ کو پر کریں۔

Q1. In a simple pendulum experiment, percentage errors (uncertainty) in "L" and "T" are 0.1% and 2% respectively. Maximum percentage error (uncertainty) in "LT^2" is:

(A) 4.1%
(B) 3%
(C) 2.1%
(D) 3.2

Q2. If $\bar{A}.\bar{B} = |\bar{A}\times\bar{B}|$ then angle "θ" between vectors $\bar{A}$ and $\bar{B}$ is:

(A) 0
(B) π/4
(C) π/2
(D) π

Q3. When a massive body of mass M_1 is moving with velocity V_1 collides with lighter stationary body of mass m_2, the velocity of massive body after collision will be:

(A) $V_1' = 2V_1$
(B) $V_2' = V_1$
(C) $V_1' = V_1$
(D) $V_2' = 2V_2$

Q4. Moment of inertia of a body spinning about its axis DOES NOT depend on the:

(A) Angular velocity of body
(B) Mass of body
(C) Orientation of the axis
(D) Distribution of mass around axis

Q5. The bodies A and B of equal masses have velocities ratio in 2:1 their K.E are in the ratio of:

(A) 2:1
(B) 1:2
(C) 4:1
(D) 1:4

Q6. The fluid is said to be incompressible, if its density is:

(A) Constant
(B) Very high
(C) Very small
(D) Zero

Q7. Equation of continuity is based upon law of conservation of:

(A) Momentum
(B) Energy
(C) Mass
(D) Charge

Q8. Glass is an example of _____ solid.

(A) Crystalline
(B) Amorphous
(C) Poly-crystalline
(D) Ductile

Q9. Young's Modulus, Y=

(A) (ΔL/L)/(F/A)
(B) (FA)/(ΔL.L)
(C) (ΔL.L)/(FA)
(D) (F/A)/(ΔL/L)

Q10. A Carnot engine undergoes _____ processes in one cycle.

(A) Two
(B) Three
(C) Four
(D) Six

Q11. Stars moving away from the earth show:

(A) Red shift
(B) Blue shift
(C) Black shift
(D) Yellow shift

Q12. A negatively charged particle is placed in a uniform electric field directed from south to north. In which direction will the particle move after being released?

(A) North
(B) South
(C) East
(D) West

Q13. A 100 ohm resistor is connected across the terminals of a 10V battery. The power dissipation of the resistor is:

(A) 1 watt
(B) 10 watts
(C) 100 watts
(D) 1000 watts

Q14. The SI unit of magnetic induction is:

(A) Nm (B) Gauss

(C) Weber (D) Tesla

Q15. Relativistic mass is always:

(A) Less than rest mass (B) Greater than rest mass

(C) Equal to rest mass (D) Zero

Q16. Inertial frame of reference is a frame in which _____ law of motion holds.

(A) 1st (B) 2nd

(C) 3rd (D) 4th

Q17. Two up quarks and one down quark make a:

(A) Meson (B) Photon

(C) Neutron (D) Proton

> Note: Important Formulae: $(K.E)_{tran} = \frac{1}{2}mv^2$ • $I = mr^2$ • $P = VI$ • $P = V^2/R$ • $\bar{A}.\bar{B} = ABCos\theta$ • $\bar{A}\times\bar{B} = ABSin\theta$ • $L = I\omega$ • $v = r\omega$

SECTIONS – B & C (Subjective Type)

Time Allowed: 2:35 Hours | **Maximum Marks:** 68 | **Paper Code:** 1HA-I 25004 (D)

Q2. (Marks (14 x 3 = 42)) Q.2 Answers the following parts briefly. (Use of graph paper is not necessary. Candidates can make their own grid on answer book if required)

(i) [03] Show that the equation $P = \rho gh$ is dimensionally homogeneous. — OR (03) — Explain that increase in entropy means degradation of energy.

(ii) [03] Time for 10 vibrations of a simple pendulum is 17.6s. Least count of stop watch is 0.1s. Find time period of simple pendulum. Also determine uncertainty in it. — OR (03) — When water falls from a tap, its cross-sectional area decreases as it comes down. Explain why?

(iii) [03] Under what condition the rectangular (perpendicular) components of a vector have same magnitude? — OR (03) — Discuss the motion of two bodies after collision when $m_1 << m_2$ and $V_2 = 0$.

(iv) [03] What is meant by Gyroscope? Explain briefly. — OR (03) — What is meant by work? How can work done be calculated graphically by a variable force?

(v) [03] A space ship has diameter of 8m rotates about its axis to produce artificial gravity of magnitude $17.47 ms^{-2}$ in it. Calculate with which angular speed the satellite rotates? — OR (03) — Differentiate between conservative and non-conservative forces. Give examples.

(vi) [03] Derive the relation between torque, moment of inertia and angular acceleration. — OR (03) — Derive the formula for Kinetic Energy [using the equations of motion]

(vii) [03] What is meant by Magnus effect? Give its causes and effects. — OR (03) — Differentiate between reversible and irreversible processes. Also give examples.

(viii) [03] Show that: Volt/meter = Newton/Coulomb — OR (03) — Find the power delivered by an engine for attaining velocity=(4î+5ĵ) m/s while it exerts a force= (8î−2ĵ) N.

(ix) [03] What is meant by null method of measurement? Enlist two applications. — OR (03) — Briefly describe the terms, stress, strain and Young's modulus.

(x) [03] Why are the suspension bridges written with their period of use? Why after that period it is dangerous to use the bridge? (Elastic potential) — OR (03) — What is meant by magnetic flux linkage? Explain briefly.

(xi) [03] State three statements of second law of thermodynamics. (one in terms of entropy) — OR (03) — How is Doppler's effect helpful in diagnosis of cardiac diseases? Explain briefly.

(xii) [03] Calculate entropy change when 800g of ice melts at 0°C, latent heat of fusion for ice is 3.36×10^5 J/kg. Also discuss sign of entropy. — OR (03) — Can sound waves be polarized? Explain briefly.

(xiii) [03] To increase intensity of waves by a factor of 25, calculate the required increase in amplitude of wave. — OR (03) — What is super fluidity? Give some of its applications.

(xiv) [03] Under what conditions can the terminal potential differences of a battery exceed its emf? — OR (03) — State the characteristics of vector product.

> Note: Attempt the following questions. (Use of graph paper is not necessary. Candidates can make their own grid on answer book if required)

Q3. (Marks 1+4+2) Q.3 What is meant by projectile? Determine the expressions for velocity and displacement of a projectile thrown horizontally from a certain height. — OR (07) — What is meant by electric field strength? Show that: $E = -\Delta V/\Delta r$

Q4. (Marks 1+2+3) Q.4 State and explain terminal velocity. Derive its equation by using stoke's law. — OR (2+4) — What is meant by quark? Explain the quark family of particles.

Q5. (Marks 1+6) Q.5 What are stationary waves? Discuss the formation of harmonics in stationary waves in an open organ pipe. — OR (1+6) — What is centripetal acceleration? Derive an expression for centripetal acceleration in terms of angular velocity "ω ".

Q6. (Marks 06) Q.6 Discuss and derive an expression for magnetic force acting per unit length of conductor, when current is passing through two parallel straight conductors in upward direction. — OR (2x3) — Discuss the following consequences of special theory of relativity: i) Time dilation ii) Length contraction iii) Mass increase

Note: Important Formulae: $\Delta S=\Delta Q/T$ • $v_2'=(2m_1/(m_1+m_2))v_1$ • T=Total time/No.of vibrations • $P=\bar{F}.\bar{v}$ • $I\propto A^2$ • $\omega=\sqrt{(g_t/R)}$ • uncertainty=Absolute uncertainty/No. of vibrations • $\Delta Q=mL_f$ • $v_1'=((m_1-m_2)/(m_1+m_2))v_1$ • $W=KE=F\Delta d$ • $F_drag=6\pi\eta rv$ • $P_1+\frac{1}{2}\rho v_1^2+\rho gh_1=P_2+\frac{1}{2}\rho v_2^2+\rho gh_2$